

Constraining the multiplicity of Y dwarfs down to planetary masses at 0.5-1000 au with JWST

Institut Trottier de recherche sur les exoplanètes
Trottier Institute for Research on Exoplanets



Université de Montréal

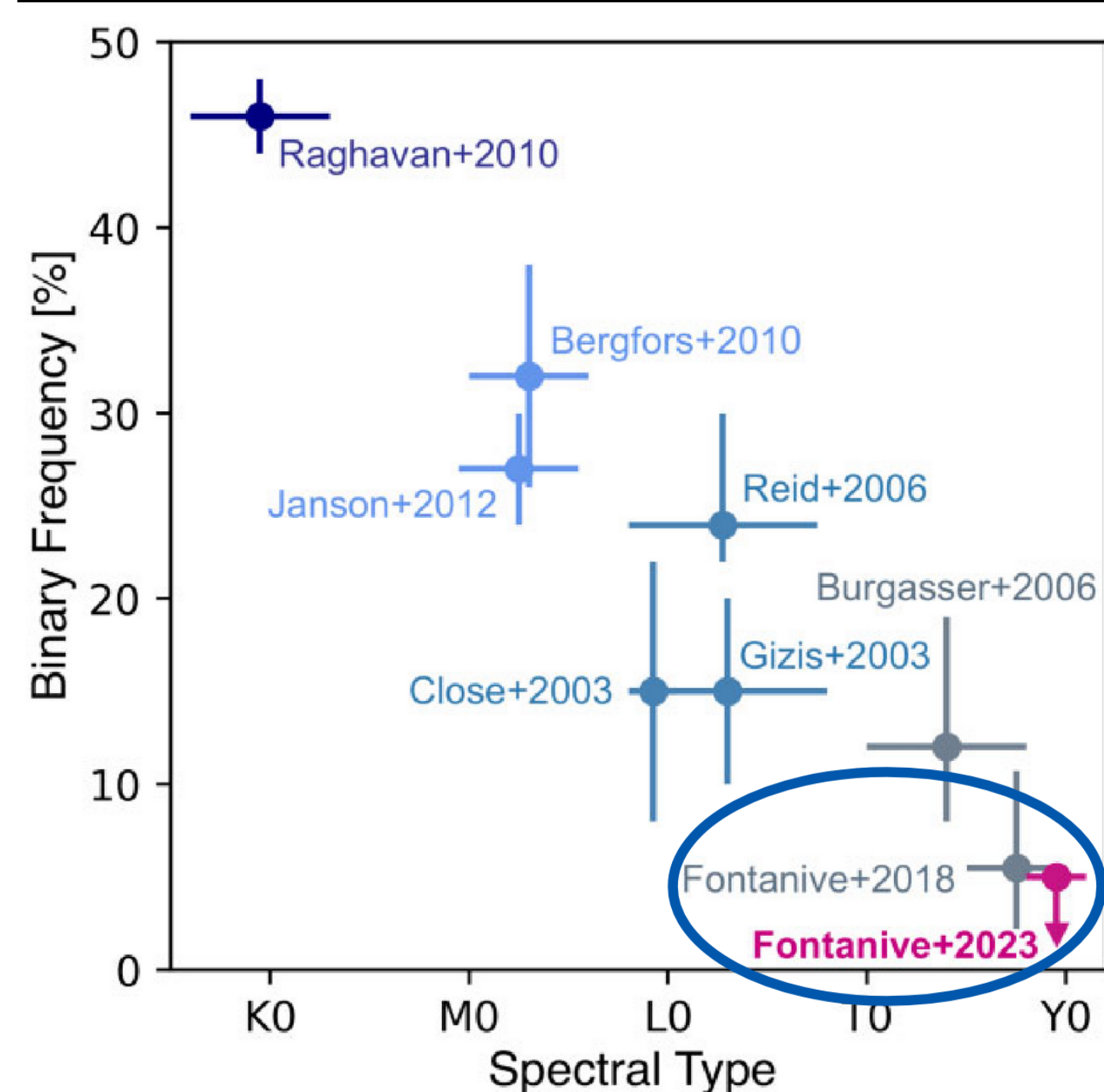
Thomas Vandal¹ C. Fontanive^{2,1} M. Giovinnazzi³ L. Albert¹
R. Doyon¹ D. Bardalez Gagliuffi³ F. Martinache⁴ D. Mary⁴
J. Kammerer⁵ P. Calissendorff⁶ M. De Furio^{7,6} M. Meyer⁶



thomas.vandal@umontreal.ca

¹Université de Montréal – Institut Trottier de Recherche sur les exoplanètes (iREx)
²Institute for Astronomy - University of Edinburgh ³Amherst College ⁴Université Côte d'Azur
⁵European Southern Observatory ⁶University of Michigan ⁷University of Texas at Austin

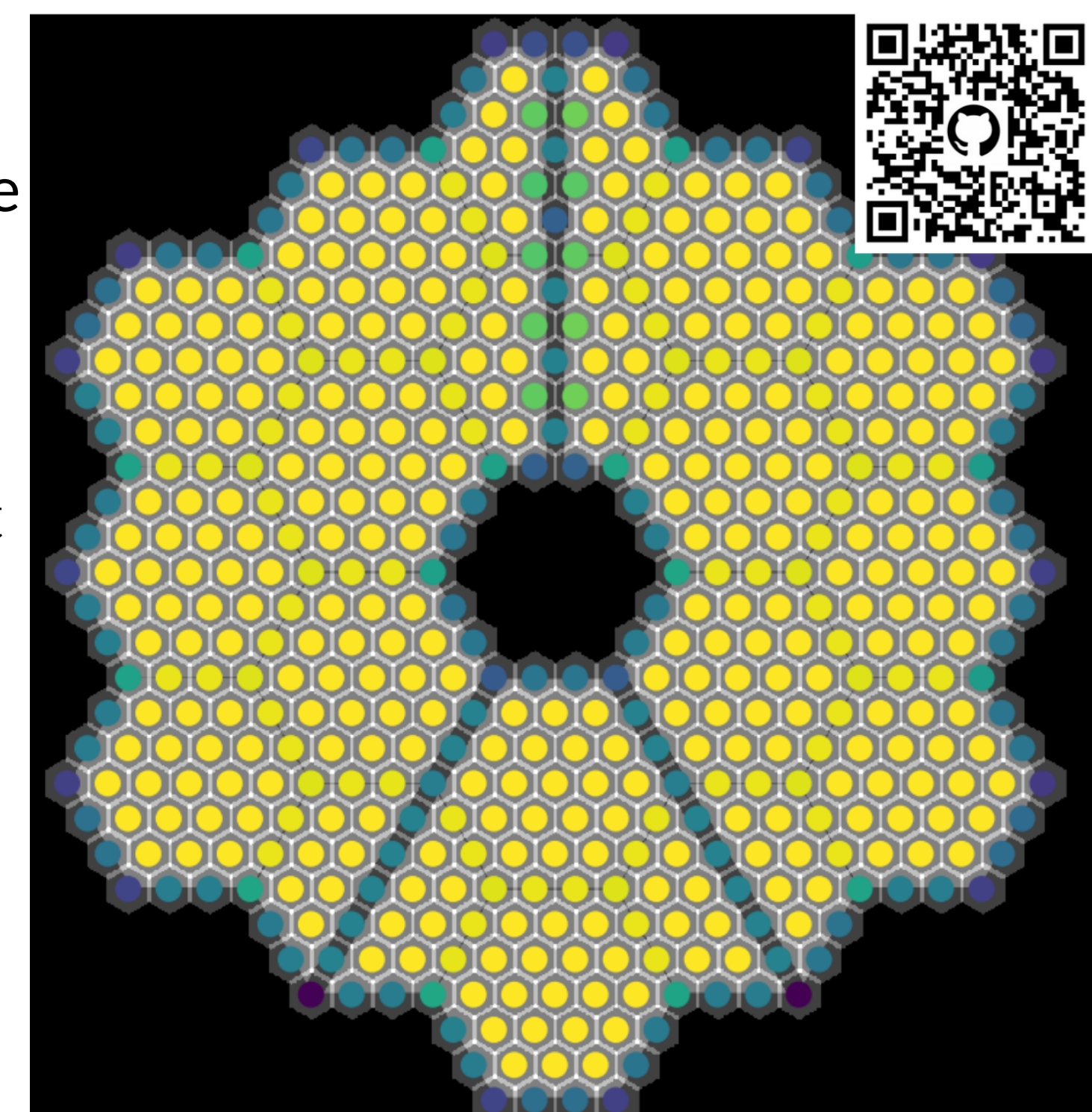
Multiplicity survey of 22 Y dwarfs with JWST



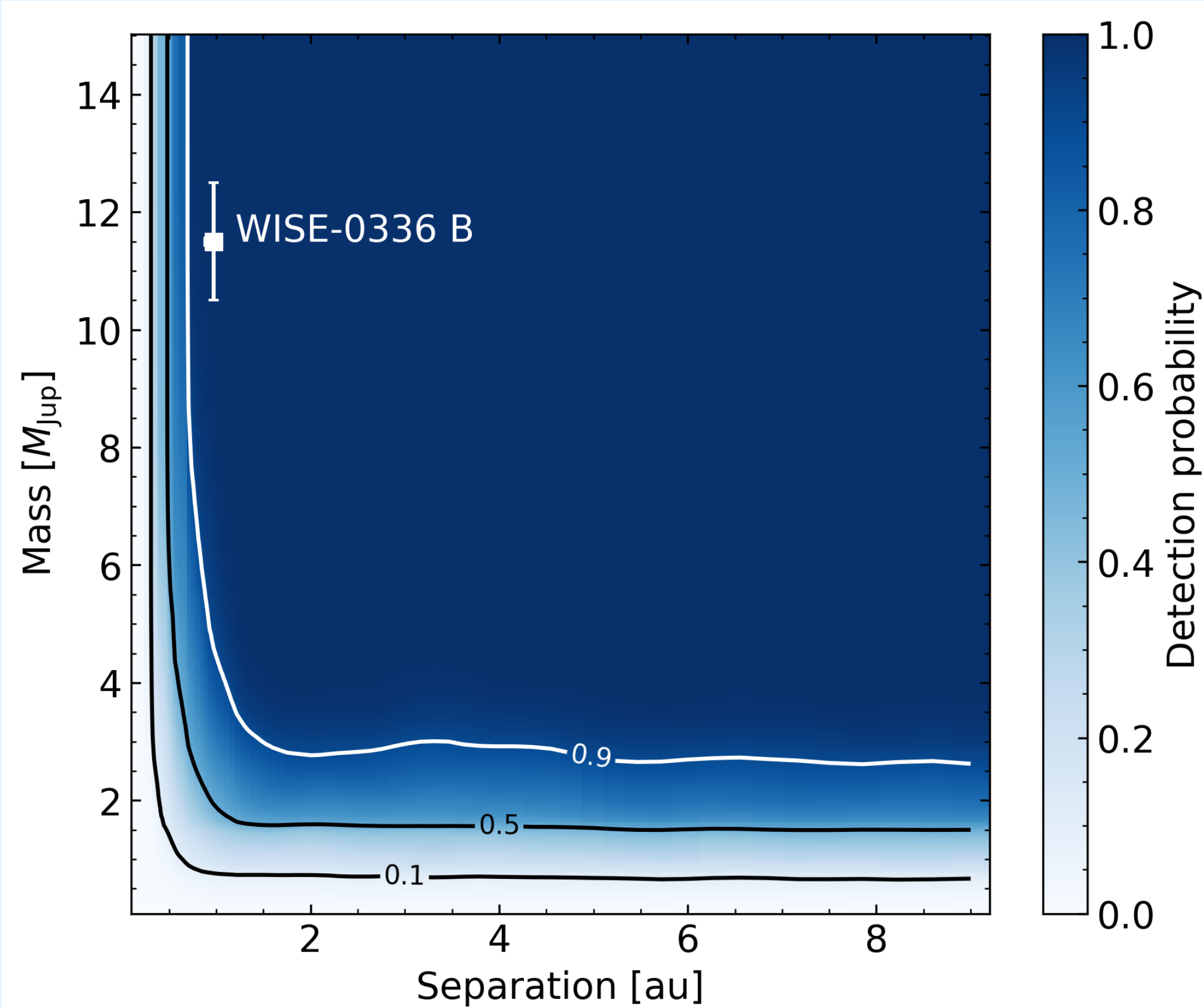
- Multiplicity can be used to distinguish stellar formation mechanisms that produce different binary fractions
- Multiplicity decreases throughout the brown dwarf sequence
- However, for Y dwarfs **HST results were limited to a few objects** and separations beyond 2 AU [1, 2]
- To confirm this trend and probe short separations, **we surveyed 22 Y dwarfs with JWST NIRCам and NIRISS at 4.8 μ m**

Kernel Phase: imaging below the diffraction limit

- We analyze the JWST images with kernel phase Imaging: we **model the telescope as an interferometer** made of small sub-apertures [3]
- This unlocks relatively high contrasts below the diffraction limit of JWST
- The pipeline is available on GitHub!
- We **recover the Y+Y dwarf binary WISE-0336 AB** with this technique and obtain parameters consistent with ePSF fitting [4]



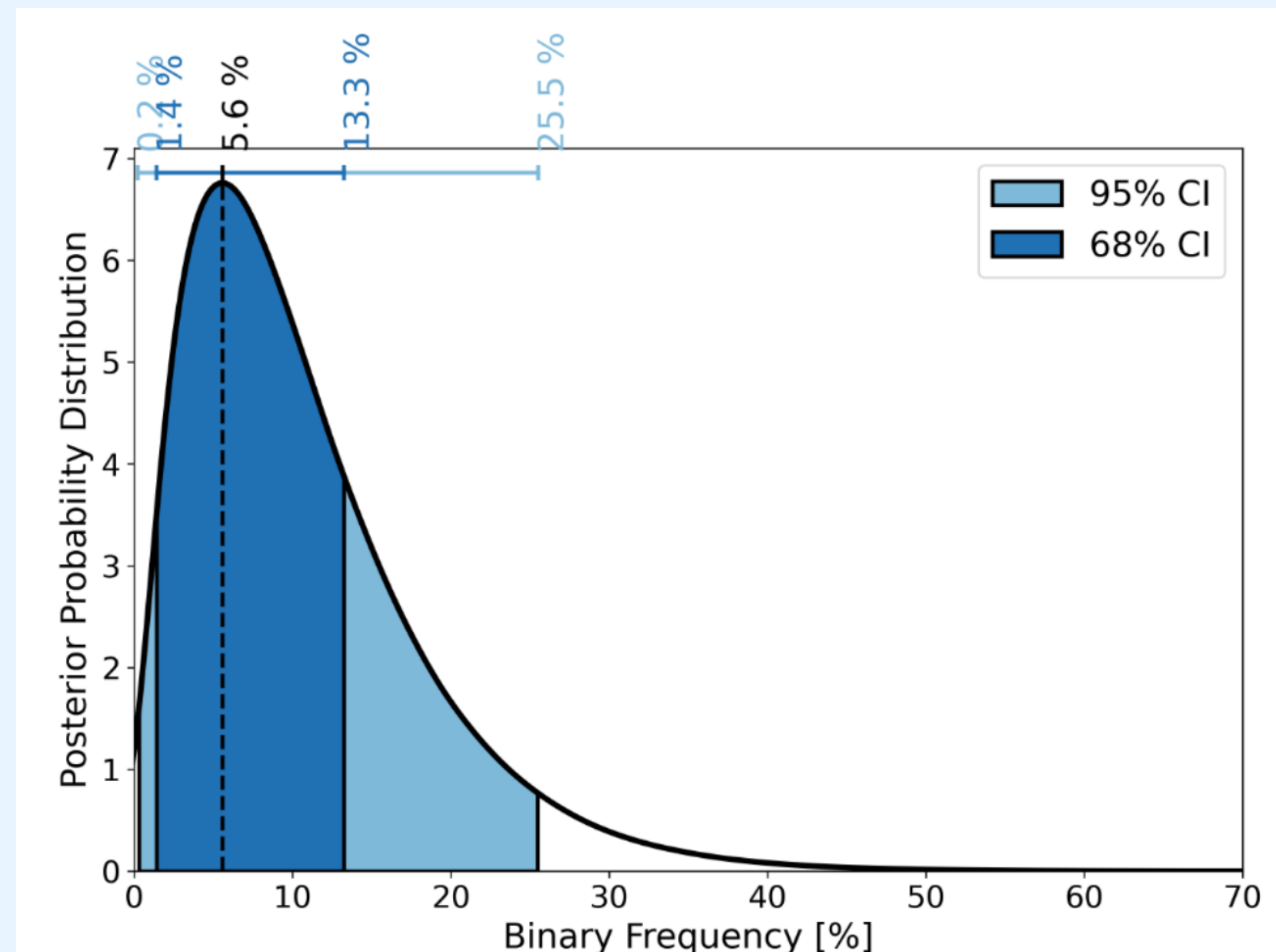
We are sensitive to companions down to 3 M_{Jup} below 2 AU with 90 % completeness and constrain Y dwarf multiplicity to $5.6^{+7.7}_{-4.2}$ %



We use our kernel phase contrast limits to derive a completeness map for the survey (left)

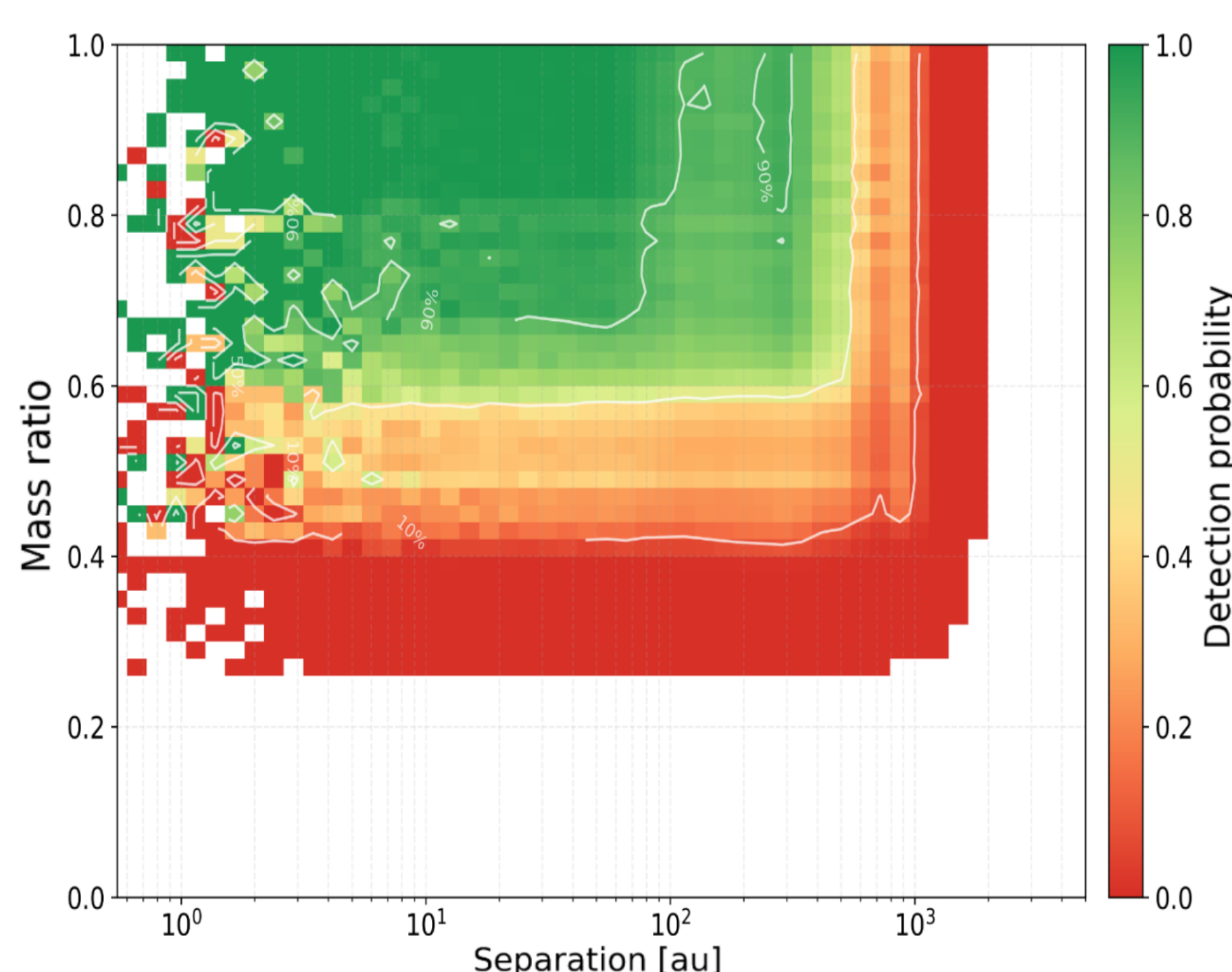
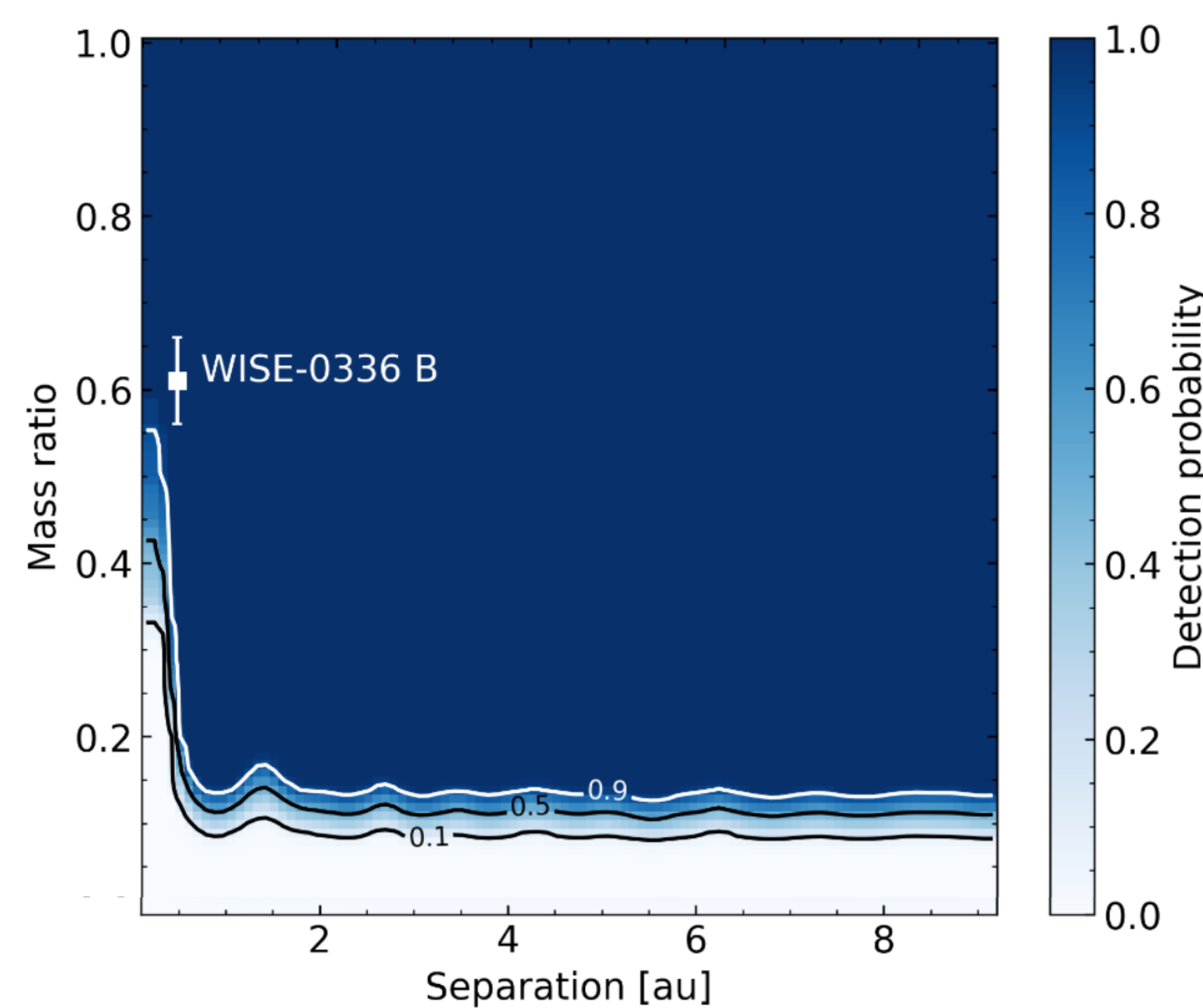


We then use this completeness map and synthetic populations to estimate the binary fraction in a Bayesian framework (right)



We also performed a wide separation search up to 1000 au

No wide companions were found. We are currently combining the completeness maps in the two regimes to derive the global completeness and binary fraction. **This is work in progress.**



Future Work

- Papers coming up: Vandal et al. (in prep), Giovinnazzi et al. (in prep)
- Detailed comparison between kernel phase and ePSF fitting
- JWST Cycle 5 Survey 10110 (P.I. Fontanive, Co-P.I. Vandal) to search for companions around more of the coolest brown dwarfs
- Follow-up WISE-0336: Track the orbit to get dynamical mass measurements

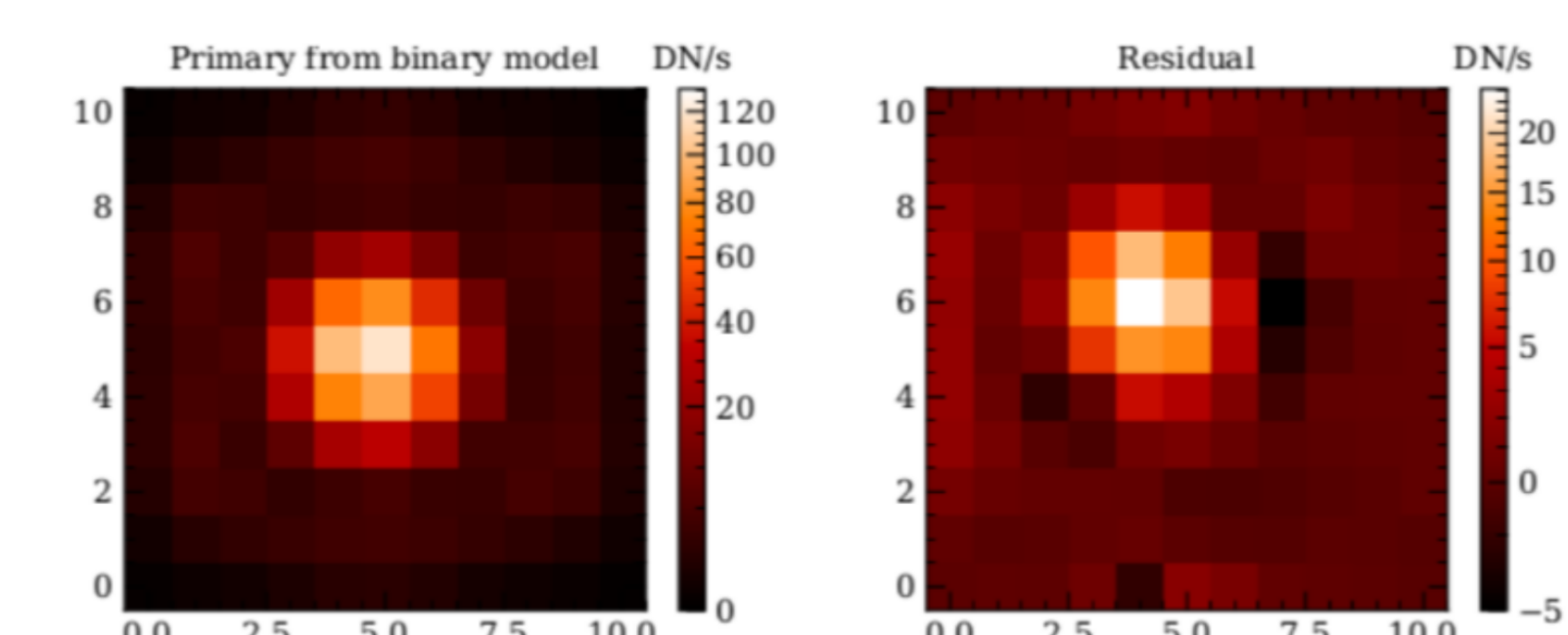


Figure showing the ePSF discovery of WISE-0336 B [4]

References

- Fontanive et al., 2018, *MNRAS*, 479, pp. 2702–2727.
- Fontanive et al., 2023, *MNRAS*, 526, pp. 1783–1798.
- Martinache, 2010, *ApJ*, 724, pp. 464–469.
- Calissendorff et al., 2023, *ApJ*, 947, p. L30.

See the PDF version of this poster!

